

ESP32 OLED Weather / System Display PCB

A custom 2-layer PCB designed in KiCad featuring an Arduino Nano ESP32, OLED display, and Bosch BME280 environmental sensor.

Features

- USB-C powered
- OLED display interface
- Temperature, humidity, and pressure sensing
- I²C communication bus
- Status LED
- Reset pushbutton
- Dual-layer ground plane
- Clean DRC (0 errors, 0 warnings)

Tools Used

- KiCad
- Arduino Nano ESP32
- SSD1306 OLED
- Bosch BME280

Skills Demonstrated

- Schematic capture
- PCB layout
- Power distribution
- I²C routing
- Ground plane design
- DRC debugging
- Hardware design workflow